

CHRIST (Deemed to be University), Bengaluru
Department of Computer Science

Project Diary

Course: CSC481-5, Computer Science Project
Semester V

ThinkStack

A Record of Decisions, Implementations and Corrections

Submitted by

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July 2026

About this Diary

This diary records the development of ThinkStack from 5 June to 7 August 2026. It is reconstructed from the repository itself: 250 commits, 59 merged pull requests, 51 recorded architecture decisions, and the published releases. It is not written from memory.

Two conventions are observed throughout. First, every quantity was measured or counted, never estimated. Second, decisions that turned out to be wrong are recorded alongside those that were right, because a diary that contains only successes teaches nothing.

The work was carried out jointly by the three team members. Entries describe what was decided and built rather than who performed each action.

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Contents

About this Diary	1
1 Timeline at a Glance	4
2 Sprint I: Building the Application	4
2.1 5 to 16 June: foundations	4
2.2 17 June: naming, audit, and the decision log	4
2.3 19 to 21 June: desktop shell and encryption	4
2.4 23 to 24 June: the paper writer	5
3 Sprint II: Making It Usable by Others	5
3.1 7 to 10 July: distribution	5
3.2 20 to 22 July: hardware awareness	5
3.3 25 July: testing and the Rust rewrite	5
3.4 27 to 29 July: first releases	6
3.5 30 July: the release that did not work	6
3.5.1 Diagnosis	6
3.5.2 What the same investigation uncovered	6
3.5.3 The correction that mattered most	7
3.6 30 July: bundling the typesetting engine	7
3.7 30 July: verified delivery	7
4 Sprint III: Correctness of What Was Delivered	7
4.1 31 July to 1 August: the pipeline, and what a green tick means	7
4.2 1 to 3 August: LitGraph	8
4.3 2 August: separating measurement from decision	8
4.4 4 August: an update mechanism that could not succeed	8
4.5 4 to 5 August: models the user chooses	9
5 Decisions and Their Consequences	10
6 What Was Learned	10
6.1 Testing source code is not testing a product	10
6.2 Suppressed errors are expensive	10
6.3 Fix categories, not instances	10
6.4 Documentation drifts silently	10
6.5 5 August: choosing a model by measuring it	11
6.6 5 August: a test that had never failed	11
6.7 6 August: a default that was indistinguishable from a decision	11
6.8 6 August: a passing suite for a feature that did not work	12
6.9 7 August: a project was always a directory	13
6.10 7 August: interface controls that a desktop application does not have	14
6.11 7 August: the constraint was packaging, not hardware	14
6.12 8 August: a check that tested for silence	15
6.13 9 August: refusing to build on a layer I could not vouch for	16
6.14 12 August: a measurement that contradicted its own test set	16
6.15 13 August: an interface reporting what it had inferred	17
6.16 14 August: a scale, and a sidebar that leaves a rail	18
6.17 14 August: replacing the interface	18
6.18 14 August: citations, and two failures that printed instead of raising	19
6.19 14 August: repairing what extraction gets wrong	20

7	Division of Work	20
8	Present State	22
8.1	Verification status by platform	22
9	Planned Work	22
9.1	Language models trained for this application	22
9.2	An interface design philosophy, and the refactor that follows it	23
9.3	A reliable paper editor	23
9.4	Further planned refactors	24
10	Closing Note	24

1 Timeline at a Glance

Date	Phase	Event
5 Jun	Sprint I	repository created
17 Jun	Sprint I	project renamed ScholarLens to ThinkStack; decision log started
19 Jun	Sprint I	desktop shell introduced
20 Jun	Sprint I	encryption implemented
23 Jun	Sprint I	LaTeX editor and auto-healing compiler
24 Jun	Sprint I	interface unified; Sprint I closes
7 Jul	Sprint II	download page
10 Jul	Sprint II	cross-platform builds and CI
21 Jul	Sprint II	hardware-aware model routing
22 Jul	Sprint II	in-application updates
25 Jul	Sprint II	test suite introduced; Rust startup diagnosis
27 Jul	Sprint II	first public release, v0.1.0
29 Jul	Sprint II	v0.1.1, then v1.0.0
30 Jul	Sprint II	v1.0.0 found not to start; root cause analysis and correction
30 Jul	Sprint II	TeX engine bundled; validated beta published
31 Jul	Sprint III	release pipeline consolidated; every dependency pinned
1 to 3 Aug	Sprint III	LitGraph: search, analysis and gaps become one canvas
1 Aug	Sprint III	stale interface after update traced to the webview cache
2 Aug	Sprint III	capability model extracted; Bench introduced
4 Aug	Sprint III	auto-update failure diagnosed; installs on a dead channel rescued
4 Aug	Sprint III	beta v1.9.0 published
4 to 5 Aug	Sprint III	model registry, task router, and Hugging Face acquisition
5 Aug	Sprint III	baseline chosen by measurement; render tests added

2 Sprint I: Building the Application

2.1 5 to 16 June: foundations

The repository was created on 5 June. The first two weeks established the project structure, an initial working demonstration, and the first iteration of the interface.

2.2 17 June: naming, audit, and the decision log

Three things happened on the same day that shaped everything afterwards.

The project was renamed from ScholarLens to ThinkStack. A backend audit corrected a set of defects found by reading rather than by testing. Most consequentially, an architecture decision log was started.

Decision recorded. Run entirely on the user’s device, with no hosted language model. The reasoning: a researcher working with unpublished results or institutionally restricted material cannot upload it, and a student project has no budget for per-call inference. This single decision determined local inference, quantised models, a frozen runtime, and eventually a bundled type-setting engine.

The log reached 32 entries by the end of the project. Its value was not apparent at the time; it became apparent during the dependency audit in the final week, when the reasoning behind earlier choices could be read rather than reconstructed.

2.3 19 to 21 June: desktop shell and encryption

The application was wrapped in a desktop shell. The shell starts the backend as a supervised child process and stops it when the window closes, so no orphaned process is left holding a port.

Encryption was implemented with Argon2id for key derivation and AES-256-GCM for the ciphertext.

Decision recorded. Use authenticated encryption rather than an unauthenticated mode. Tampering with a stored paper is then detected at decryption instead of producing plausible but corrupted output, which for research material is the difference between an error and a silent falsification.

2.4 23 to 24 June: the paper writer

The LaTeX editor was implemented, with a compiler designed to be forgiving. A small language model produces markup that is frequently almost correct, so the compiler inserts missing package declarations, wraps bare fragments into complete documents, and continues past recoverable errors. When a pass produces no PDF, the failing environment is progressively neutralised and compilation is retried, so one malformed figure does not cost the author the paper.

Sprint I closed on 24 June with every feature working from a source checkout.

3 Sprint II: Making It Usable by Others

3.1 7 to 10 July: distribution

A download page was produced, and cross-platform builds were automated. The build configuration was made data-driven, so that adding an operating system is an edit to a configuration file rather than to build logic.

3.2 20 to 22 July: hardware awareness

Model selection was made dependent on the machine rather than fixed.

Decision recorded. Size the model against *available* memory rather than installed memory. A user has a browser and an editor open. Sizing against total memory produces a process that is terminated by the operating system partway through an analysis, which is worse than producing a slightly weaker answer.

A related decision concerned graphics hardware. The diagnosis reports the GPU but enables offload only for one specific class of device.

Decision recorded. Report the GPU, but treat all but CUDA devices with sufficient memory as unusable. The inference library shipped in the installer is a CPU build, so attempting offload elsewhere fails at model load. Reporting a capability the runtime cannot use converts a working configuration into a crash.

3.3 25 July: testing and the Rust rewrite

An automated test suite was introduced. Startup diagnosis was moved from Python into the desktop shell, written in Rust, because the Python implementation imported a deep learning framework purely to determine whether a GPU existed, which cost several seconds on every launch.

Decision recorded, with a limit. Hardware diagnosis moved to Rust because it was on the startup path and slow. Model discovery was measured at approximately 9 milliseconds and was left in Python, because the same argument does not apply to code that is already fast. The principle recorded was to move work for a measured reason, not because a language seems faster.

3.4 27 to 29 July: first releases

Version 0.1.0 was published on 27 July, the first release to build on all three operating systems. Two defects had blocked it, both worth recording because neither was reproducible locally.

- The Windows build failed at model download. A JSON tool on that platform emits carriage returns, which left an invalid character on the end of a URL. The transfer tool rejected it. Linux and macOS were unaffected, so the defect existed only on the platform none of the team used daily.
- The macOS build failed at code signing. An absent optional credential was passed as an empty string, so the signing tool attempted to import an empty certificate rather than skipping signing.

Version 1.0.0 followed on 29 July, alongside a decision to reduce the installer size.

Decision recorded. Ship only the smaller model and offer the larger one with the user's consent, reusing weights the user may already have through other local tools. This removed approximately one gigabyte from every installer. Matching required a canonical naming key, because the same weights are named differently by each runtime, and comparing filenames caused the application to offer a download the user did not need.

3.5 30 July: the release that did not work

Observed. The published v1.0.0 installer, downloaded and run, displayed a loading screen indefinitely. One tester waited approximately 200 seconds. Continuous integration had been green throughout.

This is the most instructive day of the project.

3.5.1 Diagnosis

The startup path was instrumented before anything was changed, on the reasoning that the failure could not be fixed while it was invisible. The instrumentation identified the cause in a single run.

Three defects compounded:

1. The packaging framework installs the executable and its resources into different directories. The lookup searched only beside the executable, found nothing, and fell back to the development path of launching a system Python interpreter.
2. The Linux application image exports an environment variable that redirects a Python interpreter's library search into the mounted image. A frozen interpreter ignores this; a system interpreter does not, and it terminated before running any project code.
3. The loading screen retried without limit and had no error state, so a backend that died in under a second looked exactly like one that was slow.

3.5.2 What the same investigation uncovered

- The embedding weights had never been placed in the bundle. The first document ingested on an installed copy attempted a network download, in an application whose entire premise is that it needs no network. The code's own documentation stated the opposite.
- The model directory was derived from the current working directory, which the application image changes before launching. This was incorrect on every operating system for every installed copy; it worked only when launched from a developer checkout.

- The window terminated approximately one second after appearing, from heap corruption in a graphics path of the system webview. Every backend check passed while this happened, because the backend was healthy and only the window was gone.

3.5.3 The correction that mattered most

Rather than continue fixing defects individually, every external dependency was enumerated by walking all imports and every subprocess call across 51 backend modules.

Result. Exactly one unbundled runtime dependency existed: the TeX engine. Everything else was already inside the bundle. Bundling that engine closed the entire category rather than one instance of it.

The pattern the audit exposed was that the absent TeX engine, the absent embedding weights and the incorrect model path were the same defect: an assumption that a user's machine resembles a developer's.

3.6 30 July: bundling the typesetting engine

The engine was measured before it was adopted: a 58 MB binary and a package cache of approximately 47 MB, against roughly 1.1 GB of headroom under the hosting platform's asset limit.

The first attempt worked locally and failed on the Linux build server, which produced the most transferable lesson of the project.

Cause. The chosen engine version required a newer system C library than the build platform provides, so it never executed at all, failing in 15 milliseconds. The build step recorded a warning and continued, so the installer was assembled with an engine that had no package cache. The same binary would have reached users of long-term-support distributions, who would have had a paper writer that silently could not compile.

Correction. The engine version was chosen by measuring the minimum system library each release requires, rather than by taking the newest release. The selected version runs on the build platform and on older user systems, and reduced the bundled payload from 104 MB to 70 MB. Equally important, the build step now prints the engine's actual error and stops the build, because suppressing that error is what made the diagnosis expensive.

3.7 30 July: verified delivery

A validated beta was published to all three operating systems, with every automated check passing, including compilation of a PDF by the packaged application.

4 Sprint III: Correctness of What Was Delivered

Sprint II ended with software that installed. Sprint III began with the discovery that installing is not the same as remaining correct, and most of its work was caused by faults that only appear after a second release exists.

4.1 31 July to 1 August: the pipeline, and what a green tick means

Five release workflows had accumulated, differing only in what started them and how the version was derived; their build and publish halves were identical. They were consolidated into one, with the channel decided by which branch received the merge.

Two failures during this period shared a cause. A macOS launch check had been ordered before the macOS build and marked `continue-on-error`, so it reported success for a run in which it never executed. Separately, an unpinned transitive dependency updated overnight and broke all three platforms from a commit that changed no application code. Both were cases of a signal that appeared to mean verification and did not.

A third fault was subtler. Users who updated correctly continued to see the previous interface, because the embedded webview served `index.html` from its own cache, which outlives the application. The correction was to serve the entry document with `no-store` while allowing content-hashed assets to remain immutable.

4.2 1 to 3 August: LitGraph

The gap finder, search, and summarisation had been separate screens over the same corpus and much of the same computation. They were consolidated into one canvas, in which the literature is presented as a connected structure rather than as three lists: a question is asked once, the passages that answer it are shown, the papers containing them are shown in relation to one another, and the places where the literature is thin are marked on the same surface.

Analysis was moved off the request path and precomputed at ingestion, because a canvas that must summarise before it can draw is a canvas that appears broken. Selecting a node opens the paper itself, with the passages the map relied on marked in it, so that a claim on the canvas can be checked against its source without leaving the screen.

This was the largest single feature of the project and closes the consolidation that had been recorded as planned work since Sprint I.

4.3 2 August: separating measurement from decision

Hardware classification existed twice: the Rust shell measured the machine at startup, and the Python backend measured it again. The two disagreed, and the Rust answer silently won, leaving the Python path authoritative-looking and dead.

The responsibilities were separated. Rust measures; a single Python module converts those measurements into decisions: context size, offload, token budgets, and whether a document must be summarised in several passes. One consequence was immediate: graphics offload had been decided by a test for CUDA and dedicated video memory, which Apple Silicon can never satisfy despite having a usable GPU, so every Mac had been running on the processor alone.

The same reorganisation gave the interface a place to show its findings, and *Bench* was introduced beside Library, LitGraph and Scribe: what the machine can run, and what it runs it with.

4.4 4 August: an update mechanism that could not succeed

A tester reported that the update function did nothing on macOS. The investigation found three faults in sequence, each concealing the next.

The update endpoint is compiled into the application and cannot be changed afterwards. A rolling release tag had been renamed some days earlier, so every installation built during that window continued to poll an address that still existed and still advertised the build they already had. Separately, the manifest carried a pre-release version string while the application reported the plain version, and semantic versioning ranks a pre-release below its own release, so even a correctly aimed installation was told it was current.

The stranded installations could not be reached by any change to the source, because the address they poll is fixed in the binary they are running. They were recovered by publishing a manifest at the abandoned address that advertises the current version and points at the current

files, which is a redirection in everything but name. Testers on the beta channel now update in place.

The lesson generalised: the earlier failure and this one both arose from a change that did not declare what it superseded, leaving the system to infer intent from absence.

4.5 4 to 5 August: models the user chooses

Until this point the application used the model it shipped with, and one optional larger model, selected by a fixed table of filenames. A researcher who already had suitable weights, through Ollama, LM Studio, or a previous download, could not tell the application to use them.

Three modules were added. A *registry* records the models available to an installation and which task each may perform, persisted through the existing crash-safe write. A *router* answers which model serves a given task on this machine at this moment, with every dependency supplied by its caller, so that routing can be tested against fabricated hardware without a model runtime present. A *manifest*, generated during the build from the release configuration, records what that build bundled and which previously bundled models it replaces, so that exchanging the included model becomes one configuration edit rather than four coordinated source changes.

Imported models are referenced where they already are, never copied: the alternative would double the disk cost of a file measured in gigabytes on machines already selected for being constrained. A separate flag records whether the application created a file, and nothing deletes a file it did not create.

Acquisition from Hugging Face was added last, as the only part of the application that reaches the network. It searches on submission rather than on keystroke, states plainly that it is doing so, and constructs its download address from a repository identifier and a filename rather than accepting an address, since the local interface is reachable from the embedded browser.

Three defects found during this work shared a single form. A size was rounded before being compared against a memory budget, so any file below roughly five megabytes measured as zero and satisfied every constraint. A measured size of zero was treated by a default-value expression as *no measurement*, so a stale declared figure of ninety-nine gigabytes was preferred over the truth. And the memory budget itself, computed as free memory less a fixed reserve, floors at zero on a constrained machine: where zero already meant *unmeasured, therefore unconstrained*, so the machine least able to run anything was told it could run everything.

In each case a legitimate zero was indistinguishable from an absent value. The correction in each was the same: ask the module that owns the decision, which compares against real quantities, rather than testing a number for truthiness.

5 Decisions and Their Consequences

Decision	Reason	Consequence accepted
local inference only	privacy, cost, offline operation	far weaker models; seconds per response
quantised small models	fit consumer memory	reduced quality on structured output
frozen Python runtime	the user installs nothing	installers approach 1 GB
file-based vector store	no database to install	keyword index rebuilt per query
bundle only the smaller model	installer size	analysis quality depends on a consented download
bundle a typesetting engine	the writer must work with no LaTeX	70 MB added per installer
CPU-only inference build	one build behaves identically everywhere	GPU hardware unused
tags, not branches, produce releases	releases cannot happen by accident	a deliberate step is required
updates only when requested	an offline product must not contact a server unprompted	a user who never asks never updates
merges produce releases, tags record them	a stray tag must not publish to every user	releasing requires a reviewed merge
imported models referenced, never copied	a multi-gigabyte file must not cost twice the disk	the application depends on a path it does not own
task assignment chosen by the user	parameter count does not predict structured output	one extra step when adding a model
download addresses constructed, never accepted	the local interface is reachable from the embedded browser	only known hosts can be fetched from

6 What Was Learned

6.1 Testing source code is not testing a product

309 tests passed while the shipped installer could not start. The tests examined a source tree; users run an installer. The verification tiers added afterwards exercise the packaged application on each supported operating system, on machines that have no model and no typesetting distribution installed, so that a pass is evidence about a user's machine.

6.2 Suppressed errors are expensive

Two separate diagnoses were slowed by output sent to a null device. In both cases the fix was to print the error and stop, rather than warn and continue. A build that continues after a step has failed produces an artefact nobody intended to ship.

6.3 Fix categories, not instances

Three defects were the same defect. Enumerating every dependency once bounded the problem and ended it, where fixing them one at a time as testers reported them would not have converged.

6.4 Documentation drifts silently

Instructions told macOS users to use a mechanism the operating system had removed, and a LaTeX error message offered a command for one Linux distribution to every user on every platform. Both were written when they were true. Neither was noticed until a person followed them and could not proceed.

6.5 5 August: choosing a model by measuring it

The bundled model was to be replaced with something lighter and newer. Qwen3 0.6B was the obvious candidate: smaller than the model it would replace, a generation ahead, and its download verified. Every structural check passed.

It returned an empty object.

Gap finding constrains the model's output with a grammar that permits only JSON from the first token. A reasoning model begins by thinking, and the grammar leaves that nowhere to go, so it emits the shortest legal document instead: `{}`. The parser succeeds. Nothing downstream notices. The feature returns no gaps and reports no error.

A bake-off was written against the two flagship jobs, and it changed the decision entirely. The incumbent Qwen2.5 0.5B recovered four of six stated limitations, produced valid LaTeX, and was the fastest of the four candidates tested; nothing larger justified its additional weight. The model that had been about to ship on the strength of its specification came last.

One further correction belongs here, because it was the harness rather than the product. The LaTeX test initially wrapped its output in JSON to make it machine-checkable, and two candidates were marked down for producing corrupt markup: inside a JSON string, `\begin` is a valid escape and the parser silently converts it to a control character. Scribe does not do this; it asks for plain text. A test that differs from the production path measures the test.

6.6 5 August: a test that had never failed

Three times in one week the application shipped a blank page: the build succeeds, because the markup is valid, and the interface framework then throws at render time on a name that no longer exists, leaving an empty document and an error visible only in a browser console.

A test was added to mount every screen. It passed. The bug was then reproduced deliberately, the import removed again, and the test still passed, because it mounted the four page components and never the application shell, which is where the missing import was. It would have passed indefinitely while the product failed to start.

The same exercise was applied to the rest of the new work: each guarantee was broken on purpose to confirm something noticed. Two of five did not. The protection against deleting a file the application did not create had a clear error message, an obvious intent, and no test at all.

The general lesson is worth stating plainly, because this project has now learned it twice in different forms. A suite that has never failed is a hypothesis. Sprint II established that testing source code is not testing a product; this establishes that writing a test is not the same as being protected by one.

6.7 6 August: a default that was indistinguishable from a decision

Before beginning work on the paper editor, the repository was surveyed for material that no longer had a purpose. Two modules in the gap-finding package were found to be unreachable: an earlier implementation that asked the model for gaps and for research directions in two separate calls, replaced some weeks earlier by a single merged call. A configuration file for a hosting provider the project no longer deploys to was removed likewise, and a file named as though it belonged to the automated suite was renamed, since the test runner is scoped to one directory and had therefore never collected it. Roughly two hundred lines were deleted and nothing referred to them.

Reading the surviving implementation next to the one it replaced revealed something the deletion had nearly concealed. The older module named the task it was performing when it called the language model; the merged replacement did not. That argument is what selects the model. The application allows a user to assign a particular model to gap finding, and the routing

layer honours that assignment by looking up the task it is given. Given no task, the argument takes its default value of *general*, so the lookup is performed against the wrong entry, matches nothing, and falls back to the bundled model: while the interface continues to display the model the user selected. Gap finding, one of the two features the project presents as its principal contribution, had been running on the smaller model and disregarding the user's choice entirely.

The fault is the same one recorded on 4 August for the writing task, and it arrived by the same route: two calls were merged into one and the routing argument was not carried across. What makes this class of defect durable is that its symptom is indistinguishable from correct operation. Nothing fails. The output is plausible, because a smaller model still produces gaps; it produces worse ones. The only external evidence is a quality difference that a reader would attribute to the difficulty of the task.

Following the principle recorded earlier in this diary: that categories should be fixed rather than instances, the correction was accompanied by two forms of protection. The first drives each feature with the model replaced by a recording stand-in and asserts on the task the feature *requested*, rather than on the answer it received. The second parses the source of every module and fails if any call to the generation interface omits the argument at all. The second was written because the first can only cover entry points somebody remembered to enumerate.

Run once, it reported two further instances that had not been suspected: per-document analysis, merged from two passes in exactly the same way and losing the argument in exactly the same way, and metadata extraction at ingestion. The second proved not to be a fault. Extracting a title and an author list is shallow work performed on every upload, with a regular-expression fallback immediately beneath it, and directing it to the larger model would make every ingestion wait on a model exchange for no benefit. It was therefore made to state *general* explicitly. This is the more useful half of the outcome: the remedy is not that every call must name a heavier model, but that every call must record which model it intends, so that a deliberate choice and a forgotten argument stop appearing identical in the source.

The correction was verified by restoring the defect deliberately and confirming that both protections failed, then removing it again and confirming that both passed.

A feature that had been abandoned in planning was also removed in full. A conversational assistant remained present as a mounted interface with no way to reach it, and was still described as working in the user-facing documentation. Removing it exposed a dependency worth recording: the paper editor imported its retrieval step from that module, because assembling relevant excerpts within a size budget had been written there first. That step is not conversational: it takes a query and returns text, so rather than delete it, it was moved beside the search implementation it is built upon and given a name describing what it does. Deleting the feature without reading its dependents would have removed the grounding from the paper editor, which is the feature the following work was about to address.

6.8 6 August: a passing suite for a feature that did not work

The sidebar was made to withdraw when a user begins working, giving the width to whichever screen they are on, and to return when the logo is pressed. The requirement as first implemented was read as a list: opening a paper on the map, opening a document in the editor, expanding an entry in the library, and any operation on the model bench. Each of those handlers was made to ask the shell for room, and each was covered by a test.

The tests passed. The feature was reported as not working on two of the four screens. Both accounts were correct. Everything on the list behaved; everything absent from it did nothing, and most of what a person actually presses on those two screens was absent. The specification had been satisfied and the requirement had not, because the requirement was a property, any interaction, and it had been implemented as an enumeration.

The correction removed more code than it added. The shell now observes its own content region and withdraws on any press within it, so no feature calls anything, imports anything,

or can forget to. Controls nobody listed are covered, including those not yet written. Two details were necessary rather than decorative: the observation is of the press rather than the click, because selecting a region of the map is a drag that may never produce a click; and it is taken during the capture phase, because several rows deliberately stop propagation on their own buttons and a listener waiting for the event to bubble would never be told.

A second distinction was needed once actions could close the sidebar. Its state had been remembered between runs, correctly, since a person who collapses it has expressed a preference. An automatic collapse writing to that same record would mean opening a single paper teaches the application to start collapsed permanently, a setting the user never chose, altered by an action unrelated to settings, with nothing on screen to explain it. The deliberate and the automatic are now separate, only the first is remembered, and the logo clears both, since clearing only its own would leave it apparently dead while a feature still held the other.

The lesson is a narrower form of one this diary has already recorded twice. It has been noted that a suite which has never failed is a hypothesis, and that categories should be corrected rather than instances. This adds that a test can fail in neither of those ways and still be worthless: it can verify precisely what was implemented while the implementation answers a different question from the one asked. Tests written from a list inherit the list's omissions, and they inherit them silently, since a case nobody thought of is a case nobody writes an assertion for.

6.9 7 August: a project was always a directory

A user reported that a colleague's paper would not compile. The engine said it could not load a figure, and then that it had divided by zero. The second message was the first one's consequence: the graphics package had read a width from a file it never opened.

The cause was not in the compiler. A paper had been stored as a directory since the first version, and compilation had always run with that directory as its working directory, so the relative path in the document was correct. What was missing was any way to put a second file into the folder. The document was the only file the application could reach, so a figure could be referenced and never supplied.

The correction was to expose the directory that already existed: list, read, write, upload, rename, move, copy and delete, presented as a file tree with the papers themselves as its top level. The row of chips that had previously selected a paper was removed: it was a second picker for the upper level of the same hierarchy, and there had been no way at all to reach the level beneath it.

The security work is the part worth recording. The interface reaching this API is a web view, which the desktop framework treats as remote content, so a filename arriving here is untrusted input in the same sense that a downloaded address is: `../../../ssh/id_rsa` is a filename. Every operation therefore resolves its argument against the project directory and refuses anything that lands outside. Resolution happens *before* the check rather than after, which is what catches a symbolic link: nothing about the name `notes.tex` suggests it points at the system password file, so an inspection of the string would pass it through. Deleting the containment check was tried deliberately, and the tests fail.

A second failure arrived the same day from the same paper. Documents with an index compiled to `Undefined control sequence \indexentry`. An index requires two passes with an external step between them: the first pass writes a list of entries, a program converts that list into a formatted index, and the second pass includes it. The bundled engine performs the equivalent step for bibliographies automatically but not for indexes, so the intermediate file was never converted and the index package fell back to reading the raw list as if it were markup.

The obvious remedy was to call the standard index program. It is present on the developer's machine and would not be present in the installer, which is the same error as assuming a system typesetting installation: precisely what bundling an engine was meant to eliminate. The index is therefore generated by the application itself, which is a modest amount of code for a small

and stable format, and is consistent with the compiler's existing habit of repairing what it is given rather than refusing it.

6.10 7 August: interface controls that a desktop application does not have

Three menu items, new file, new folder, new paper, were reported as doing nothing at all. They had been built on the web browser's prompt dialog, which the desktop web view does not implement; it returns nothing, and the code read that as the user having cancelled. Deletion had the same dependency on the browser's confirmation dialog, where the failure would have been worse: a guard written as "if not confirmed, stop" either prevents the action permanently or, read the other way, performs it without ever asking.

Naming now happens in a row within the tree itself, where the file will appear, and destructive actions use a dialog belonging to the application. That dialog was made shared rather than local, because it was about to be the third independent implementation in the codebase; three dialogs is three sets of keyboard handling, focus behaviour and dismissal rules to get right, and the safest button is the one that does nothing, which is where focus now begins.

Behind that fault was a second one that would have prevented the menu from working even with a functioning dialog. The menu dismissed itself on any press elsewhere, and listened for that press during the capture phase: necessary, because much of the tree stops events propagating. It did not check whether the press was inside the menu, so pressing an item removed the button before the click could reach it. Neither defect was visible from the code; both required the application to be run.

The recurring lesson is the one this project has now recorded in several forms. The suite was green, the components rendered, and the feature did not exist. What was missing was not a test but an execution: none of this can be discovered without launching the thing a user launches.

6.11 7 August: the constraint was packaging, not hardware

A tester reported that the application told him his graphics card could not be used. The message was accurate. The inference engine is shipped compiled for the processor only, so the card genuinely could not be used, and the diagnosis had correctly said so. What the message did not do was offer anything, explain why, or indicate whether it would ever change. A correct sentence at a dead end.

The first design took the obvious route. The manufacturer of that card publishes prebuilt libraries, so the plan was to download those on request. It was implemented to the point of working before a more careful look at the numbers undermined it. Those libraries are approximately four hundred megabytes, and they depend on a further half a gigabyte of mathematical libraries that are present only on machines where a developer toolkit has been installed. Close to a gigabyte, for one manufacturer's hardware.

The alternative had been overlooked because nobody had packaged it. The same inference library can be built against a vendor-neutral graphics interface whose runtime component is installed alongside every graphics driver. Querying that interface on the development laptop, a machine with none of the manufacturer's software installed, and which had been recorded in these notes as having an unusable card, returned three devices, the integrated processor graphics, the discrete card through an open-source driver, and a software renderer. Two of the three were immediately usable. The proprietary route would have reached neither.

The download for that approach is about ninety megabytes and requires nothing else, because the driver component is already present on any machine that can display graphics at all. The trade is genuine: the vendor-neutral path is slower on that vendor's hardware than the proprietary one. It is far faster than the processor on every machine, which is the comparison that matters when the alternative is no acceleration at all.

The pattern is worth recording because this is the third time it has appeared. The language model is not included in the installer because of a file size limit imposed by the distribution platform. Typesetting worked everywhere only once the typesetting engine was included rather than assumed. And graphics hardware was unusable because of which build was shipped. In each case the question looked like “is this machine capable” and the answer was “yes, but can it be delivered”. Capability was never the constraint; packaging was.

Two errors were made during the work and are worth recording as much as the result. The first: the software renderer must be excluded. It is the processor presenting itself as a graphics device, so directing work to it would route through a translation layer to reach the hardware already performing the calculation, while reporting success at every step. The second: the memory each device reports is not its own. Both real devices claimed twelve gigabytes, which is the system’s memory rather than the card’s four, and the software renderer claimed the most of all. An early version ranked devices by reported memory, which would have selected precisely the device that must never be selected. Both were found by running the detection and reading the output, not by reasoning about it.

A third error was caught by the supervisor rather than by any test. The module that answers hardware questions is deliberately constructed so that every fact is supplied to it rather than discovered by it, which is what allows a fabricated machine to be reasoned about on hardware that resembles it in no way. A convenience function was added that consulted the real graphics interface from inside that module. It was noticed immediately, from the outside, by someone asking whether that module was not supposed to be self-contained. It was, and the addition had quietly reintroduced exactly the class of assumption the module had been written to eliminate.

6.12 8 August: a check that tested for silence

Three failures in one day turned out to share a shape, and the shape is worth recording because it is not obvious until the third one.

The graphics engine described in the previous entry was finally built and published. It failed twice before it succeeded, for unrelated reasons, and then failed a third time against its own verification step. That step asserted that the compiled library reported support for offloading work to the graphics processor. It reported that it did not. The library was correct: ninety megabytes of compiled shader code, exactly as intended. The function being asked, however, reports whether a graphics backend has *registered at runtime*, not whether one was *compiled in*, and the machine performing the build has the development libraries but no graphics driver. The assertion was testing the build machine rather than the artefact.

The same day, a release built correctly, uploaded every file, reported every stage successful, and remained unpublished. An unpublished release is invisible; requests for its files return “not found”; and the update mechanism treats “not found” as “you are already up to date”. Testers were therefore told they were current while the release sat unreachable. The instruction to publish had been given. It had simply not been obeyed, and nothing checked.

The third instance was the most instructive because it is still in the application. Metadata extraction falls back to the language model when the faster method fails, and the condition guarding that fallback tests whether the extracted title is *empty*. The faster method almost never returns an empty title. It returns a wrong one. The better path is therefore unreachable, not because it is broken, but because the broken path always reports success.

The common form: a guard that checks for absence, protecting against a failure mode that is confident wrongness. Absence is easy to detect and rare. Confident wrongness is common and looks identical to success. Where a value can be legitimately empty, zero, false, an empty string, a check for emptiness cannot distinguish “no answer” from “a bad answer”, and the second is the one that reaches the user.

6.13 9 August: refusing to build on a layer I could not vouch for

The next feature was citation support: allowing an author to cite a paper from their own collection, which the application can do and comparable tools cannot, because it has already read the paper. The entry it would generate is built from metadata extracted at ingestion, and that extraction was written by a collaborator. Rather than build on it, I asked to see it work.

The instrument was a short script that follows one document through every stage and preserves the output of each, so the intermediate results could be examined rather than described. Three published papers of established standing were used, so that correct answers were known in advance.

Not one of the three produced a usable author list. On the best known of them the extracted title was the copyright notice printed above it; on the others the title was correct but had authors' names attached to the end of it. In every case the author list contained institutions, the employers named beneath the authors, and in two cases the paper's own title appeared as an author. The publication year, taken as the first four-digit number appearing near the start of the document, was wrong for the most significant of the three.

The cause is that the extractor receives the document as unstructured text. A document of this kind is not text; it is positioned glyphs, each carrying a size and an orientation, and the title of a paper is reliably the largest horizontal text near the top of the first page. That information exists in the file and is discarded before extraction begins. Restoring it, with two exceptions for a rotated identifier stamp printed down the margin of one archive's documents, identified all three titles exactly.

Two things are worth separating here. The first is that the fault was not one of implementation quality but of information: a correct implementation of the wrong question. The second is that I had considered rewriting the component in a systems language for reliability, and measurement showed that the entire stage accounts for under one percent of processing time, the remainder is the embedding model, which is already compiled native code. A rewrite would have reproduced the same wrong answer faster.

This metadata labels every node of the literature map. The map has therefore been displaying copyright notices where paper titles were intended, in released software, for some weeks. It was not reported by anyone.

6.14 12 August: a measurement that contradicted its own test set

The repair followed from the diagnosis. Extraction was rebuilt to read the positioned glyphs rather than the flattened text, separated into two components: one concerned only with page geometry, which groups glyphs into lines and divides each line at its column gaps, and one concerned with what a title or an author is. The separation was not tidiness. Three of the four most damaging faults were geometric and had nothing to do with scholarly documents: lines grouped by the top of their bounding box rather than by the baseline they share, which split a title set in small capitals into single letters; a change of font within a word, which inserted a space into it; and an accent stored as a separate glyph before its letter, which left a surname unsearchable. None of these are visible while reading a procedure that is simultaneously deciding whether something is a name.

Removing institutions from author lists was treated as a question of structure rather than of vocabulary. A list of employers is never complete and dates immediately. Three rules were used instead: a structural word such as *university* or *institute* disqualifies an entire line rather than a single entry, since a line of affiliations may name only some of them in terms a list would recognise; a short capitalised abbreviation standing beside ordinary words denotes an organisation; and a phrase printed more than once within the author region is an address rather than a person, because a shared address is printed once per author while a name is printed once per document. The third rule requires no vocabulary of any kind, and is the only one capable

of distinguishing a country from a surname, the two being indistinguishable by spelling.

The result on the documents used during development was complete: every title, year and author list exact. That number was then discovered to be worthless. A second measurement was constructed in which neither the documents nor the expected answers were chosen by the author: papers were sampled across fifteen subject classifications through a public archive interface, and the archive's own catalogue records were taken as the expected values. Accuracy on that sample was sixty-two percent, against one hundred on the development set.

The entire difference lay in typesetting conventions that had not been examined. One physics template spaces a centred author line widely enough that the spacing is indistinguishable from column separation, so a line of two authors was read as five fragments and none of them as a name. Another runs from the affiliations directly into the abstract without printing a heading, so the search for authors continued into the body of the paper and collected section headings. A margin threshold, introduced to discard an archive's identifier stamp, was discarding the first line of any centred title wide enough to begin left of it. Correcting these raised the sample to ninety-three percent of titles and eighty-six percent of author lists exactly correct.

The lesson is narrower than it first appears. The development set had not been chosen dishonestly; it had been chosen from a single field, and a template is a convention of a field. A measurement taken over material one has selected oneself describes the selection, not the method. The remaining failures, mathematics within a title, publisher cover pages preceding the document, and submissions too recent to carry an identifier: are a long tail of the same kind, and the established solution to a long tail of layout conventions is a classifier trained over layout features rather than a further accumulation of rules. That is what the mature tool in this area does, and it is the point at which this approach should be expected to stop improving.

6.15 13 August: an interface reporting what it had inferred

The collection screen is one member's work throughout, and it had been extended with panels summarising the state of the application: which model was in use, how many analyses had run, what was being written. Reviewing the layers beneath those panels found that two of the four figures had never displayed a value in any release, they were fixed placeholders, and that a third was inferred rather than read.

The inference is the instructive one. The panel naming the model in use searched the list of installed models for the first whose status was *ready*. No such status exists; the application reports *present*. The search therefore never matched, and the panel silently displayed whichever model happened to be first in the list. Beneath that lay a larger error: there is no single model in use. Work is routed by task, and which model serves a task depends on every other model installed: what it has been assigned, how much memory is free at that moment, and precedence between them. The backend computes that and records the reason in a comment written weeks earlier: duplicating the rule in the interface would let the two drift. On the development machine the larger model that actually performs analysis did not appear in the list being searched at all. The panel now reads the routing table the backend already produces.

A related fault ran through the same screen, a single dash stood for three different facts, not yet loaded, nothing to report, and the request failed. An outage was therefore indistinguishable from an empty library. This is the same shape as the extraction guard described above, and the third instance recorded this month of one value being asked to carry several meanings.

Two further findings concerned the layout rather than the data. Both of the pane-based screens sized themselves by subtracting a constant from the height of the window, with a comment estimating the height of the page heading. That is a measurement replaced by a guess: when the headings were removed the constant was wrong, silently, and the panes stopped reaching the bottom of the window. Replacing it with a layout that fills the space it is given failed on the first attempt, because the property used to do so resolves against the nearest enclosing flexible container and an unclassed wrapper sat between the two. Both screens then collapsed

to the height of their own contents. The correction was to pass the height explicitly at every level.

The lesson recorded is not about the technique. A mechanism was replaced without first establishing what it sat inside; the original was fragile, but it worked, and the replacement was correct in principle and wrong in that structure. The same mistake recurred twice more in the same session, filling a column whose neighbour then grew taller, and equalising panels whose contents were not equal: before the layout was expressed as flow rather than as position.

6.16 14 August: a scale, and a sidebar that leaves a rail

The author of the collection screen judged that it presented too much at once, and reduced it: six summary figures that had opened the page were moved to caption the section they describe, two were removed outright, and the document list was divided into pages. The removals are the interesting part. One reported the number of bytes the stored papers occupy: accurate, cheap to compute, and of no use to any reader, since nobody is short of the space in question and the figure changes nothing about what they do next. The other reported how many fragments the text had been divided into for indexing, which describes the implementation rather than the library. A figure earns its place by answering a question the reader would actually ask, and neither did.

Two faults found during that review belonged to the shell rather than to the screen.

The first was that the interface had twelve different small type sizes, 0.68, 0.70, 0.72, 0.75, 0.76, 0.78, 0.80, 0.82, 0.84, 0.85, 0.90 and 0.95 of the root size: distributed across a hundred and sixty-one declarations in two stylesheets. Differences of two hundredths are not decisions; they are what accumulates when each component chooses independently, and the effect is an interface that reads as careless without any single value being identifiably wrong. These were consolidated to six named steps. The root size was also made to vary with the width of the window rather than being fixed, since the application is read at arm's length on displays between thirteen and twenty-seven inches. A first attempt allowed too wide a range and produced text that read as magnified rather than as legible; the range was narrowed.

The second concerned the navigation panel, which collapsed to nothing. That sounds like the greatest possible gain of space and is the reverse, the two controls still required when it is closed, the mark that reopens it, and the button explaining the current screen, then have to be drawn on top of the page, obscuring whatever is beneath them. Collapsing the panel to get it out of the way therefore put two things in the way. It now collapses to a narrow rail that carries both, and because the page's margin is derived from the same declared width, the content moves aside for the rail rather than passing under it.

One defect from that change is worth recording for its shape. Moving the explanatory button onto the rail made it vanish. The panel is drawn above it in the stacking order, so the button was rendered behind the rail: present, correctly positioned, and invisible. Nothing in the change suggested a stacking question, and nothing in the source indicated one; it was found by looking at the screen and observing that the control was not there. It is a reminder that a layout change is verified by looking, not by reasoning about what should follow.

6.17 14 August: replacing the interface

The interface was replaced. The original had reached a point where the same class of correction kept recurring: a dozen font sizes chosen independently by different components, a navigation panel that collapsed to nothing and therefore had to draw its remaining controls on top of the page, and panes sized by subtracting an estimated constant from the height of the window. The decision taken was to replace it rather than continue correcting it.

All three members worked on the replacement. The design was built as a parallel tree while the original continued to ship, so that neither the released application nor the work in progress

blocked the other.

The arrangement has one failure mode, and it governed how the work was done: two trees drift, and the one nobody is looking at rots. Three measures were taken against it. Both trees were run through the same continuous integration matrix, so neither could reach a release having been checked differently. The test that verifies the interface calls only routes the backend defines was found to name the old directory explicitly, which would have meant that on the day the replacement shipped it went on passing while examining a directory nobody built; it now takes the directory as a parameter. And a third check was added that calls every route against a running server, because parsing both sides proves the paths agree while only calling them proves anything executes.

Every correction made during the fortnight both trees existed was made twice. That is the true cost of replacing a running interface, and it is not the rewrite but the period of double maintenance around it. Copying between the trees was not available: the replacement uses the token names the original used for sizes to name its colours instead, so a mechanical transfer would have produced a stylesheet in which a colour answered to a size.

One defect from that period is worth recording for its shape. Moving the button that explains the current screen onto the collapsed rail made it disappear. It was present, correctly positioned and invisible, because the panel is drawn above it in the stacking order. Nothing in the change suggested a stacking question and nothing in the source indicated one; it was found by looking at the screen and observing that the control was not there.

6.18 14 August: citations, and two failures that printed instead of raising

The paper editor could compile a document but could not cite one, which for a research tool is the omission that matters most. The interaction chosen was deliberately not a dialog: an author is already typing when they decide to cite, so typing the word `cite` opens the library beneath the cursor, further typing filters it, and pressing Enter replaces the word with a citation command. Choosing nothing leaves the word where it was, and an ordinary word compiles, so nothing is committed by opening the list.

One planned subsystem was removed before it was built. The intention had been to track citation numbers so that the numbering stayed correct as citations moved. BibTeX already rennumbers every citation on each recompile and orders the reference list according to the chosen style, so the entire mechanism was available at the cost of emitting keys instead of numbers. It is worth recording as a decision because the alternative would have been a second, worse copy of something the toolchain performs correctly.

Two defects were found underneath, and they share a shape with the extraction defect recorded on 12 August. Neither raised an error.

The first was in storage. Author lists were flattened into the vector store by joining them with commas, because the store holds only strings. In a bibliography the comma is not a separator between people, it separates a surname from a given name, so a list of eight authors read back as sixteen fragments of names. Nothing failed. A bibliography does not validate its input; it typesets it. The list is now encoded as structured text, and the older form is still read so that a library already built does not have to be rebuilt.

The second was that a document containing citations but no bibliography command produced neither the references nor an error. The command that prints a reference list is also the command that instructs the compiler to run the bibliography processor at all, so its absence removed both the output and the process that would have produced it. Each citation was typeset as a question mark. The compile step now supplies the command when a document cites something and the project has a bibliography file to point at, both conditions being necessary: naming a file that does not exist converts a working compile into a failing one.

A third defect is recorded for the method rather than the cause. The citation list did not appear. Every automated test passed, including tests that typed into an editor and asserted that

the list opened, because those tests run in a simulated document object model that computes no layout. Driving the actual application and reading the element's position back showed that the list was present, populated with every paper in the library, and positioned 1382 pixels down a window 1000 pixels tall, inside a region that hides rather than scrolls whatever exceeds it. A second cause was found the same way: the compiled document displayed beside the editor was drawn above the list, concealing the column that held the citation keys.

The lesson is narrow and worth stating exactly. A test environment that omits layout cannot fail a layout defect. A feature correct in every respect except where it is drawn is, to the person using it, a feature that was never built. What this means for the interface suite is set out in the testing document.

6.19 14 August: repairing what extraction gets wrong

Extraction identifies titles correctly for about 93 percent of papers and author lists for about 86 percent. Roughly one paper in seven is therefore stored under something wrong, and that record names the paper on the literature map, in every search result, and in every reference generated from it. A citation feature built on top of it inherits the error.

The question was where the correction should be made, and the two candidates were not equivalent. Editing a document's bibliography repairs that document; the next one is generated from the same wrong record and repeats the mistake. Editing the library record repairs the source, and every citation, map node and search result made afterwards is correct. The library was chosen for that reason, and the editor was deliberately left without a second facility of its own, since a project's bibliography is already a file in its tree and can be opened like any other.

One detail is worth recording because it was nearly got wrong twice in the same day. The editing form separates author names on commas, but accepts semicolons instead when any are present, because a name written surname first contains a comma. Splitting on it is precisely the defect the morning had been spent removing from the storage layer.

7 Division of Work

Work was divided by feature rather than by layer, so that each member owned a capability end to end: its data model, its interface, and its failure modes rather than a horizontal slice of everybody's. The consequence is that no part of the product can be described without naming the person who built it.

Capability	Principal author	Scope
Retrieval and knowledge base	Aditya	document ingestion, chunking, embedding index, semantic search
Analysis	Aditya	summarisation, claim extraction, theme clustering, and the precomputation that moved them off the request path
Gap finder	Aditya	gap detection over the analysed corpus, and the suggestion engine above it
LitGraph	Aditya	the canvas consolidating search, analysis and gaps; the reader, passage marking and node interaction
Scribe	Rithesh	LaTeX editor, the auto-healing compiler, and the natural language to markup path
Desktop shell	Rithesh	the Tauri application, native hardware diagnosis, process supervision, in-application updates
Delivery engineering	Rithesh	cross-platform build, release pipeline, artefact validation, and the automated test suite
Model management	Rithesh	hardware capability model, task routing, the model registry, and Hugging Face acquisition
Library	Jitvan	the paper collection screen in full, the document list, its expansion and deletion, the encryption controls, and the overview panels reporting collection size, the models in use and the drafts in progress
Encryption	Jitvan	at-rest encryption and decryption of stored documents, with the key handling and interface around it
Distribution site	Jitvan	the public landing and download page
Interface refinement	Jitvan	typography and identity, dark mode correction, and the assistant panel

The boundaries were not absolute. The interface was worked on by all three, and several features crossed hands: encryption was introduced by Jitvan and later extended when the key derivation was hardened; the desktop shell was begun as packaging work and grew into hardware diagnosis, which then fed the model routing. Where a capability changed owner, the original author's design was retained and the reasoning recorded in the decision log rather than replaced silently.

Review was mutual and enforced by the branch rules described above: no change reached the release branches without passing another member's review and the automated gate, including changes by the member who wrote those rules.

8 Present State

Measure	Value
Commits	336
Pull requests merged	82
Application code (Python)	14 438 lines, 83 modules
Test code	8 375 lines, 46 modules, 1 042 tests
Interface tests	213 tests, including every screen mounted
Interface code	13 930 lines, 23 components
Desktop shell (Rust)	777 lines
Build and release tooling	4 484 lines, 22 scripts, 7 workflows
REST endpoints	63
Architecture decisions recorded	78
Releases published	12, most recently v2.1.15
Platforms built and validated automatically	3

Figures are counted on the working branch at 14 August and include the citation work described above, which is complete and tested but not yet released. The totals are of tracked files only: generated output, dependencies and the local evaluation scripts are excluded, so the counts describe what the team wrote rather than what the repository contains.

8.1 Verification status by platform

Platform	Status
Linux	automatically validated, and the replaced interface, the paper editor and citations exercised by hand on physical hardware
Windows	automatically validated; confirmed working by a human tester before the interface was replaced, and not driven by a person since
macOS	automatically validated; exercised by a tester before the interface was replaced, and not driven by a person since. Unsigned applications are still obstructed on first launch
In-application update	working; testers update in place from the beta channel

9 Planned Work

The following are planned rather than aspirational. Each follows from a limitation identified during development, and several are already partly prepared for in the existing structure of the system.

9.1 Language models trained for this application

The clearest quality limitation is the capability of a general-purpose 0.5 billion parameter model on structured output. Asked to plot a function it produced a reference to an image file that does not exist, where a 1.5 billion parameter model produced a correct plot. The intended response is models suited to the task rather than simply larger ones, in three stages of increasing ambition.

Stage one: fine-tuned task models. Training pairs of natural language instruction and compilable LaTeX are already collected passively during normal use, and the routing table in the inference client already contains entries for a dedicated writing model, so adopting one is a matter of supplying weights. A second model fine-tuned for the structured summarisation and claim extraction formats the application parses would address the cases where a general model returns prose where JSON was requested.

Stage two: adapter distribution. Rather than shipping whole models, distribute small adapters applied over the bundled base model, so a capability upgrade costs tens of megabytes instead of gigabytes. Selection would use the hardware profile the application already computes at startup, so a machine receives only variants it can run.

Stage three: models trained from scratch. The longer term intention is to train small language models specifically for academic literature work, rather than adapting general-purpose ones. A model trained from the outset on research text, structured extraction and typesetting markup can be far smaller than a general model of equivalent usefulness for these tasks, which matters directly for a product constrained to consumer hardware.

Federated learning. A possible direction, subject to careful design, is improving these models from real usage without collecting anyone's documents. Training would occur on the user's own machine and only model updates, never text, would leave the device, and only with explicit consent. This is recorded as a possibility rather than a commitment: it must not be allowed to weaken the privacy guarantee that is the reason the product exists, and that constraint takes precedence over the capability.

9.2 An interface design philosophy, and the refactor that follows it

The interface was built feature by feature under time pressure. It is coherent but was not designed as a system: styling has accumulated into a single large stylesheet with two competing styling mechanisms in use.

The intention is deliberately ordered. A design philosophy will first be agreed by the team through discussion, and the interface will then be refactored to it. Refactoring before that agreement would simply produce a second set of accumulated decisions. Points to be settled include:

- **Design tokens.** Colour, spacing and typography expressed as variables, so the interface compiles from a defined system rather than from decisions made one screen at a time.
- **One environment rather than four.** Browsing, searching, interpreting and writing are currently separate screens. The intended direction is a single workspace whose density and visual quiet change with the task.
- **A single cross-platform identity** rather than per-platform conventions, so the application looks deliberate on every operating system.
- **Latency as a design material.** Local inference takes seconds and that cannot be removed. The interface should be designed around honest waiting rather than concealing it.
- **Reading comfort.** Sessions are long and the content is dense, so typography, measure and contrast are functional requirements rather than decoration.

9.3 A reliable paper editor

The paper writer is the most used feature and has received the most correction, but its editing surface remains a plain text area over LaTeX source. The planned work is a genuinely capable editor:

- syntax awareness, bracket matching, and compiler diagnostics shown against the line that caused them;
- a visual editing mode in which document structure can be edited without the author reading markup, with the compiled PDF remaining the single source of truth for appearance;

- position synchronisation between the source and the compiled document, so that selecting a place in one moves to it in the other;
- continued strengthening of the natural language to LaTeX path, which currently rewrites a selected passage in place and is the feature the planned writing model would most improve.

9.4 Further planned refactors

- **Incremental keyword index**, removing the per-query rebuild that currently limits corpus size.
- **Stylesheet consolidation**, resolving the two styling mechanisms into one, carried out as part of the interface refactor above rather than separately.
- **Code signing and notarisation** for macOS and Windows. This is required before a production release: both operating systems currently obstruct the first launch of an unsigned application, which every user encounters and which reads as the software being broken.
- **Bundled model integrity checks**, so that a corrupted or partial payload is detected at startup rather than at first use.

10 Closing Note

The technically demanding parts of this project, namely local inference with hardware-aware model selection, hybrid retrieval, an auto-healing typesetting pipeline and authenticated encryption, were completed and work. What determined whether anyone else could use the software was packaging, dependency completeness, and verification of the delivered artefact.

The distinction between software that works and software that has been shown to work on a machine other than its author's is the lesson this project taught most clearly, and it was taught by a release that failed rather than by one that succeeded.